

STRUCTURE OF THE SYLLABUS FOR 4 YEAR UG PROGRAMME**SCHOOL NAME** - Royal School of Biosciences**DEPARTMENT NAME** - Biochemistry**PROGRAMME NAME** - UG Biochemistry

1ST SEMESTER					
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-C
Major (Core)	BCH152M101	Biomolecules	100	3	3-0-0
Major	BCH152M111	Biomolecules and Cell Biology Practical	100	3	0-0-6
Minor	BCH152N101	Nutritional Biochemistry	100	3	3-0-0
Interdisciplinary (IDC)	IKS992K201	IKS-1	100	3	3-0-0
Ability Enhancement course (AEC)	AEC982A101	Communicative English-I	100	1	1-0-0
AEC	BHS982A101	Behavioral Science I	100	1	1-0-0
Skill Enhancement Course (SEC)	BCH152S111	Biochemical Assessment of Food Products	100	3	0-0-6
Value Added Course (VAC)		To be chosen from basket	100	3	3-0-0
*MOOCs	MOOCs 1	*MOOCs/online course will be identified by the dept. from the list of courses available on the MOOC online platform/SWAYAM portal	100	3	
TOTAL CREDIT FOR 1st SEMESTER				23	

2ND SEMESTER					
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-C
Major (Core)	BCH152M201	Proteins and Enzymes	100	3	3-0-0
Major (Core)	BCH152M213	Protein and Enzymes Practical	100	3	0-0-6
Minor	BCH152N201	Introduction to Cytology	100	3	3-0-0
IDC	IKS992K202	IKS-2		3	3-0-0
AEC	AEC982A201	Communicative English-II	100	1	1-0-0
AEC	BHS982A201	Behavioral Science II	100	1	1-0-0
SEC	BCH152S211	Food Adulteration	100	3	0-0-6
VAC		To be chosen from Basket		3	3-0-0
*MOOCs		*MOOCs/online course will be identified by the dept. from the list of courses available on the MOOC online platform/SWAYAM portal	100	3	
TOTAL CREDIT FOR 2nd SEMESTER				23	
3RD SEMESTER					
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-C
Major (Core)	BCH152M301	Metabolism of Biomolecules	200	4	3-1-0
Major (Core)	BCH152M313	Metabolism Practical	200	4	0-0-6
Minor	BCH152N301	Clinical Biochemistry	200	4	3-0-0
IDC	BCH152I301	Immunity and Health	200	3	3-0-0

AEC	AEC982A301	Communicative English-III	200	1	1-0-0
AEC	BHS982A301	Behavioral Science III	200	1	1-0-0
SEC	BCH152S311	Working with proteins	200	3	0-0-6
*MOOCs		*MOOCs/online course will be identified by the dept. from the list of courses available on the MOOC online platform/SWAYAM portal	100	3	
TOTAL CREDIT FOR 3rd SEMESTER				23	
	4th SEMESTER				
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
Major (Core)	BCH152M401	Concepts of Genetics	200	4	3-1-0
Major (Core)	BCH152M402	Human diseases and Basic Ayurveda	200	4	3-1-0
Major (Core)	BCH152M413	Practical-IV (Genetics)	200	4	0-0-6
Minor	BCH152N401	Microbiology	200	3	3-0-0
Minor	BCH152N402	Biochemical Correlation of Diseases	200	3	3-0-0
AEC	AEC982A401	Communicative English-IV	200	1	1-0-0
AEC	BHS982A401	Behavioral Science IV	200	1	1-0-0
*MOOCs		*MOOCs/online course will be identified by the dept. from the list of courses available on the MOOC online platform/SWAYAM portal	100	3	

TOTAL CREDIT FOR 4th SEMESTER					23	
	5th SEMESTER					
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P	
Major (Core)	BCH152M501	Molecular Biology	300	4	3-1-0	
Major (Core)	BCH152M502	Microbiology	300	4	3-1-0	
Major (Core)	BCH152M513	Molecular Biology and Microbiology Practical	300	4	0-0-8	
Minor	BCH152N501	Environmental Biochemistry	300	4	3-1-0	
Internship	BCH152M522	Internship		4		
TOTAL CREDIT FOR 5th SEMESTER					20	
	6th SEMESTER					
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P	
Major (Core)	BCH152M601	Advanced Cell Biology	300	4	3-1-0	
Major (Core)	BCH152M602	Immunology	300	4	3-1-0	
Major (Core)	BCH152M603	Membrane Biology and Bioenergetics	300	4	3-1-0	
Major (Core)	BCH152M613	Immunology and Cell Biology Practical	300	4	0-0-8	
Minor	BCH152N601	Basics of Genetic Engineering	200	4	3-1-0	
TOTAL CREDIT FOR 6th SEMESTER					20	
	7th SEMESTER					
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P	
Major (Core)	BCH152M701	Genetic Engineering and Biotechnology	400	4	3-1-0	
Major (Core)	BCH152M702	Research Methodology,	400	4	3-1-0	

		Bioethics, and Scientific Writing			
Major (Core)	BCH152M703	Plant Biochemistry	400	4	3-1-0
Major (Core)	BCH152M713	Plant Biology and Genetic Engineering Practical	400	4	3-1-0
Minor	BCH152N701	Cancer Biology	400	4	3-1-0
TOTAL CREDIT FOR 7th SEMESTER				20	
	8th SEMESTER				
COMPONENT	COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
Major (Core)	BCH152M801	Bioanalytical techniques	400	4	3-1-0
Minor	BCH152N801	Microbial Ecology	400	4	3-1-0
		Project / Dissertation		12	
		Or			
		Virology		4	3-1-0
		Neurobiochemistry		4	3-1-0
		Molecular Endocrinology		4	3-1-0
TOTAL CREDIT FOR 8th SEMESTER				20	